

Text-to-Mask Vision

Open-vocabulary object detection and segmentation
with Grounding DINO + SAM

image + text prompt -> boxes -> masks -> demo

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GitHub: github.com/Therad445/text-to-mask-vision



1. Why this project matters

Classical CV pipelines are powerful, but their fixed labels make fast prototyping awkward.

Fixed-class detector

- predefined categories
- retraining for new concepts
- limited interactive control

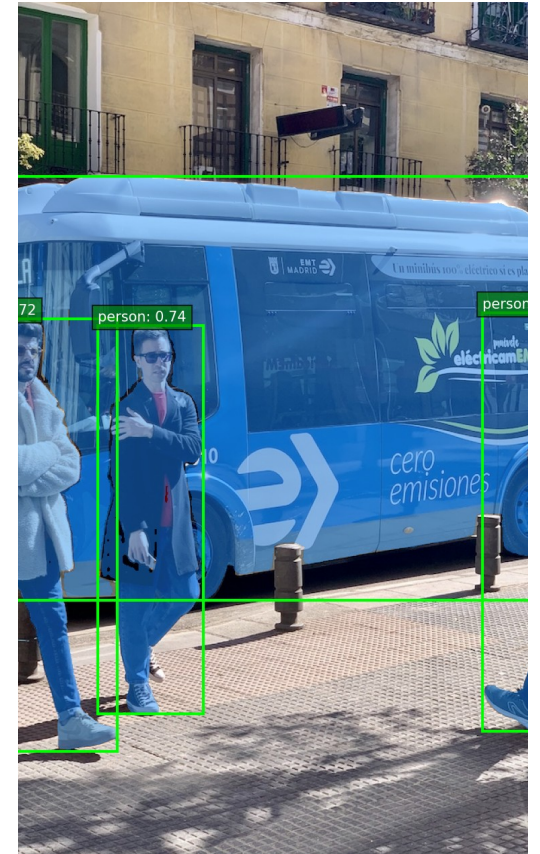
Example: model knows COCO labels, not arbitrary user intent.



Text-to-mask

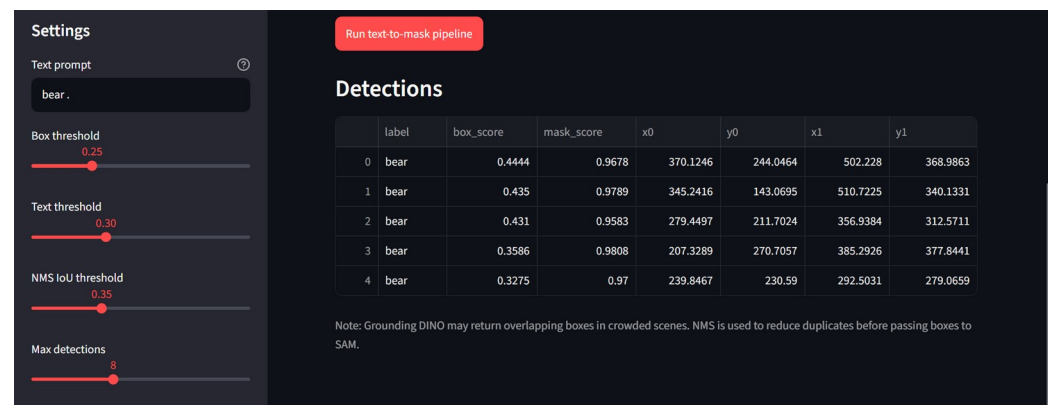
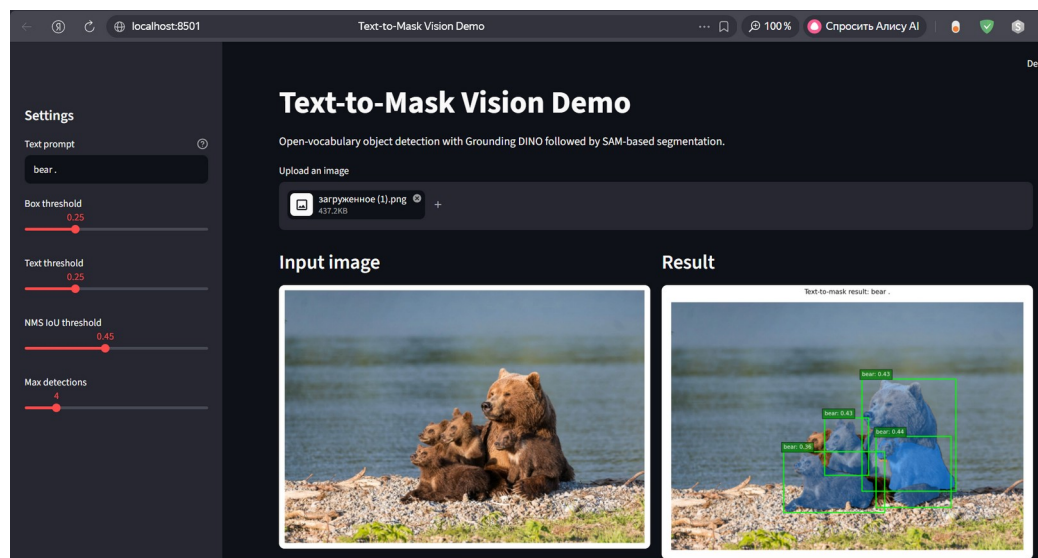
- natural-language prompt
- open-vocabulary detection
- segmentation without training

Useful for annotation, visual search, and rapid CV prototypes.



2. End-to-end pipeline

A practical chain: language-guided localization first, high-quality masks second.



Design choice: detection quality is the bottleneck; SAM follows the boxes it receives.

3. Models used

The project uses pretrained foundation models; the contribution is integration, tooling, evaluation and analysis.

Grounding DINO

- open-vocabulary object detection
- input: image + text prompt
- output: boxes, scores, labels
- Hugging Face: IDEA-Research/grounding-dino-base

localization

Segment Anything (SAM)

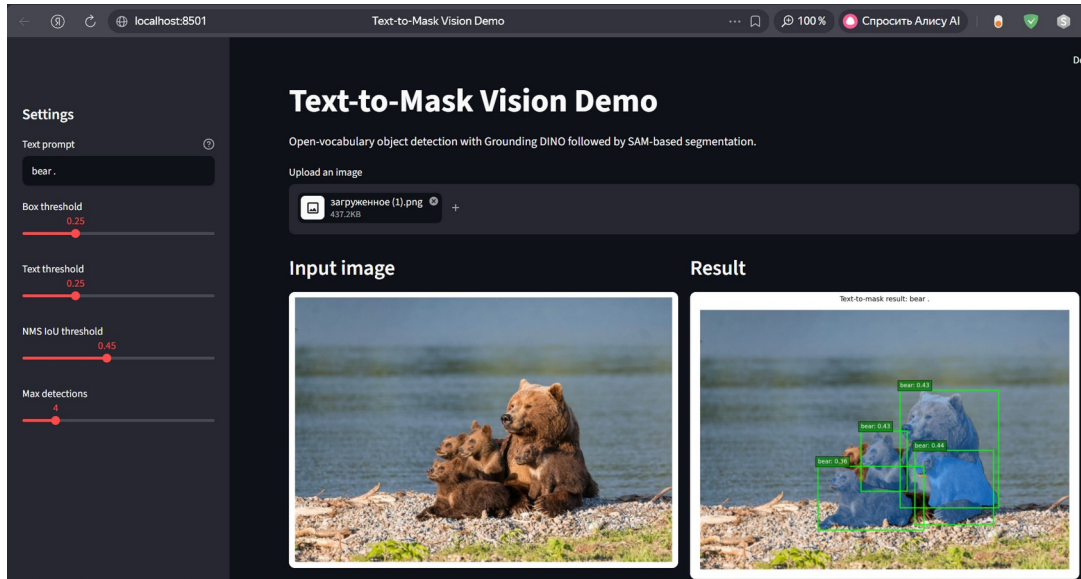
- box-guided segmentation
- input: image + selected boxes
- output: masks + mask scores
- checkpoint: SAM ViT-B

segmentation

No training from scratch - the project demonstrates how to combine pretrained models into a working CV application.

4. Implementation and repository

A reproducible project structure: app, reusable pipeline, notebooks, report, figures and checks.



src/text_to_mask_pipeline.py

reusable inference module

app.py

Streamlit demo

notebooks/02_*.ipynb

sanity check

notebooks/03_*.ipynb

batch gallery

report/final_report.pdf

final report

scripts/check_project.sh

sanity checks

Large model checkpoints are excluded from Git and downloaded separately.

PASS

clean project check

3 blocks

experiments

5 cases

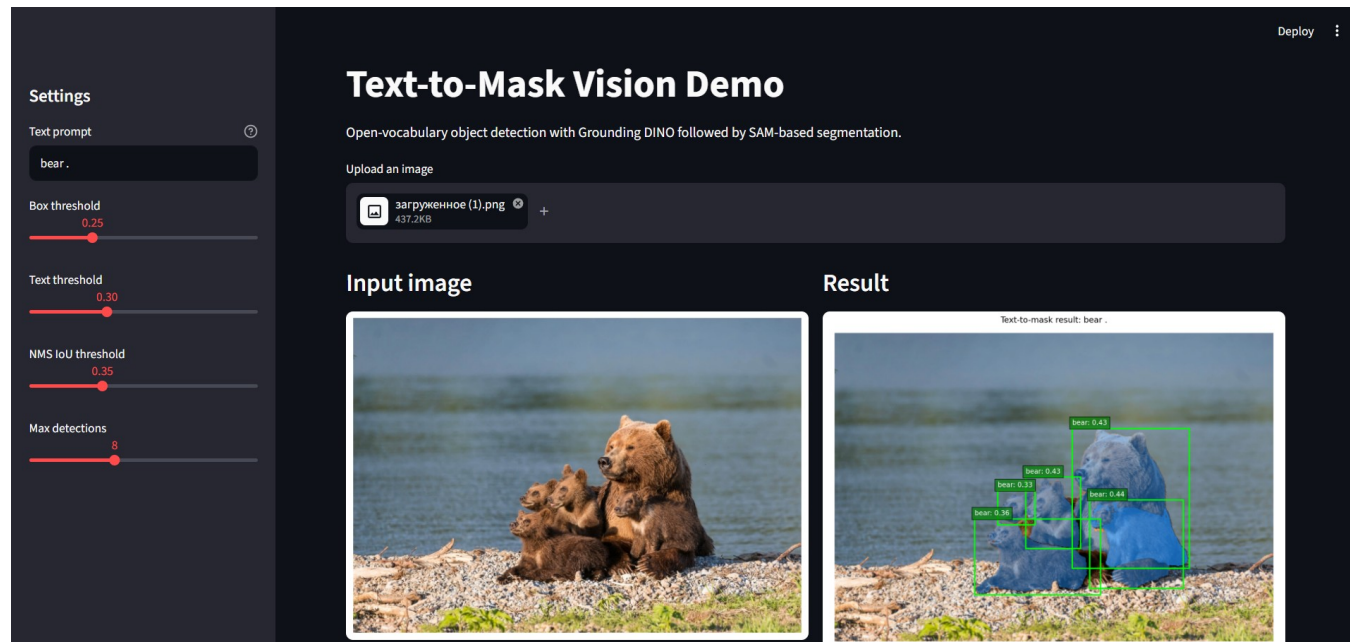
gallery cases

PDF + PPTX

final artifacts

5. Interactive demo

The system is not only a notebook experiment - it runs as a local web application.



Demo controls

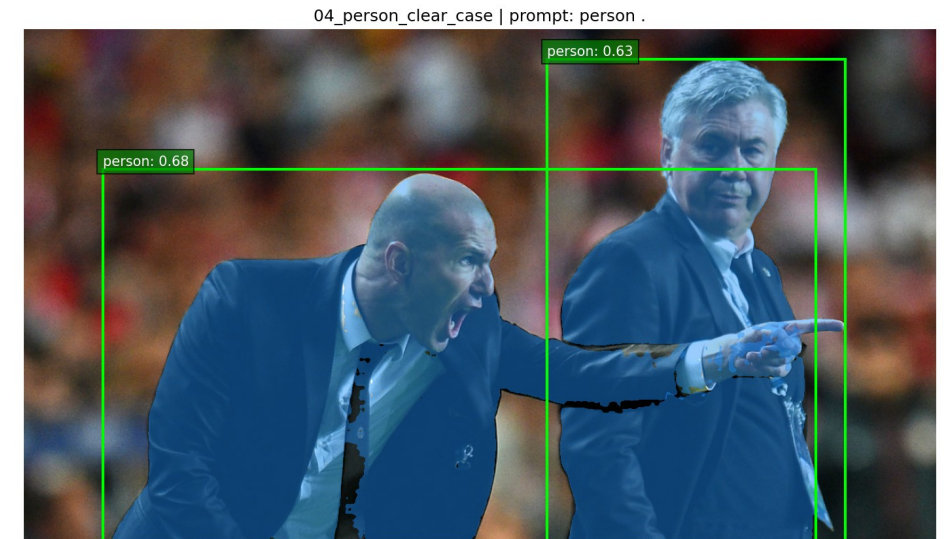
- image upload
- text prompt
- box/text thresholds
- NMS IoU threshold
- maximum detections
- visualization + detections table

Run: `python -m streamlit run app.py`

6. Experiment design

The gallery intentionally combines successes, prompt sensitivity and failure-analysis cases.

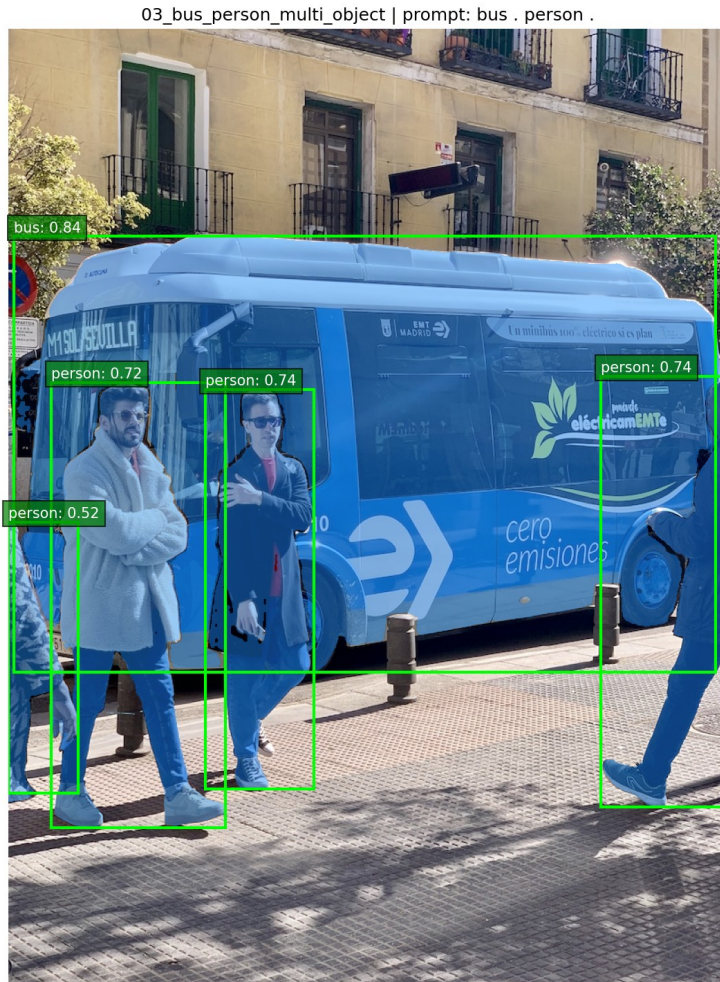
Case	Prompt	Purpose
Crowded bears	bear .	overlap / failure analysis
Bear cub prompt	bear cub .	prompt sensitivity
Bus and people	bus . person .	multi-object prompt
Clear person	person .	simple success case
Weak prompt	small animal .	mismatched prompt



Evaluation method: qualitative inspection + proxy metrics from the pipeline.

7. Success case: multi-object prompt

A single prompt can request multiple categories in the same image.



Prompt: bus . person .

- bus detected with high confidence
- several person instances detected
- masks are produced from selected boxes
- good example of open-vocabulary control

0.709

mean box score

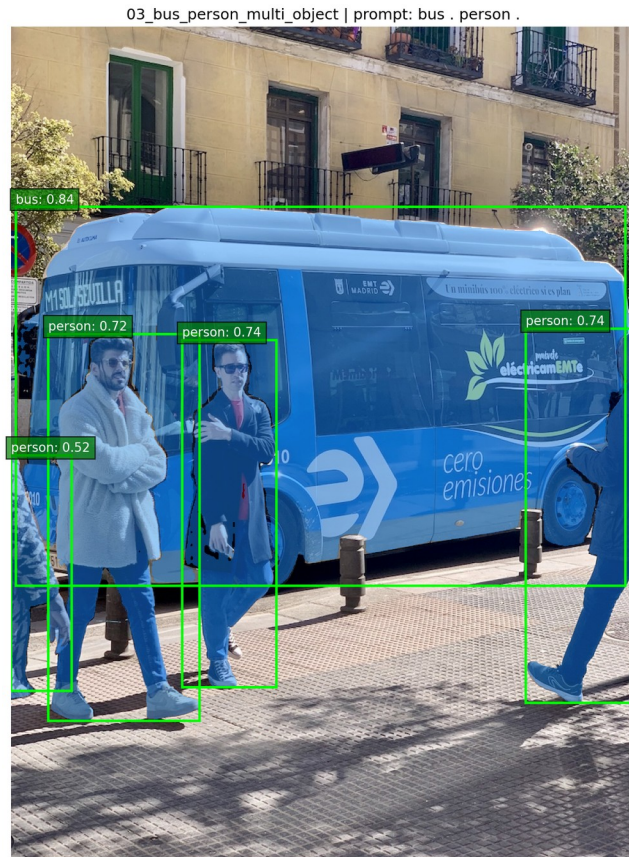
0.962

mean mask score

8. Prompt sensitivity: controlled comparison

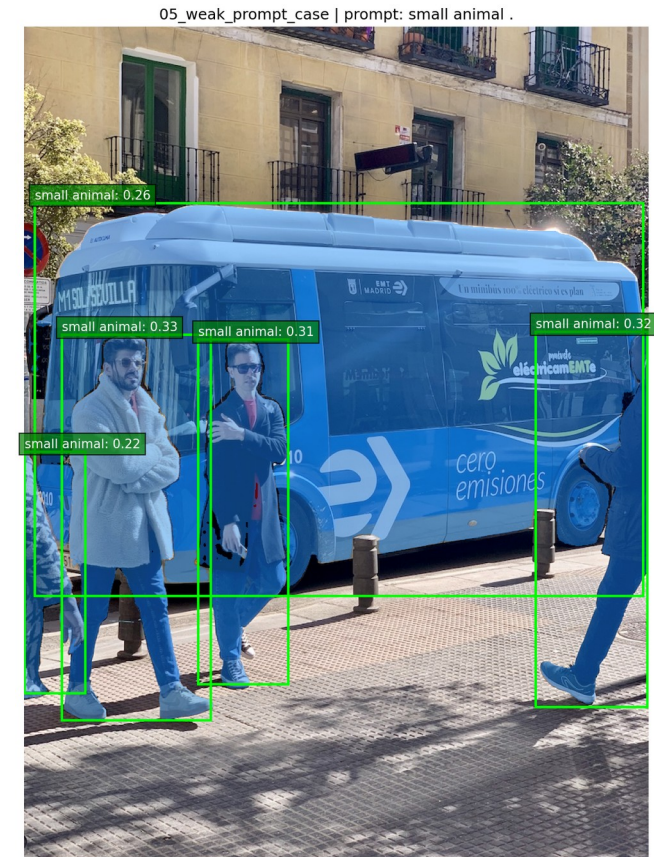
Same image, different prompts - the result changes semantically.

Semantically correct prompt



`bus . person .` -> meaningful detections

Mismatched prompt

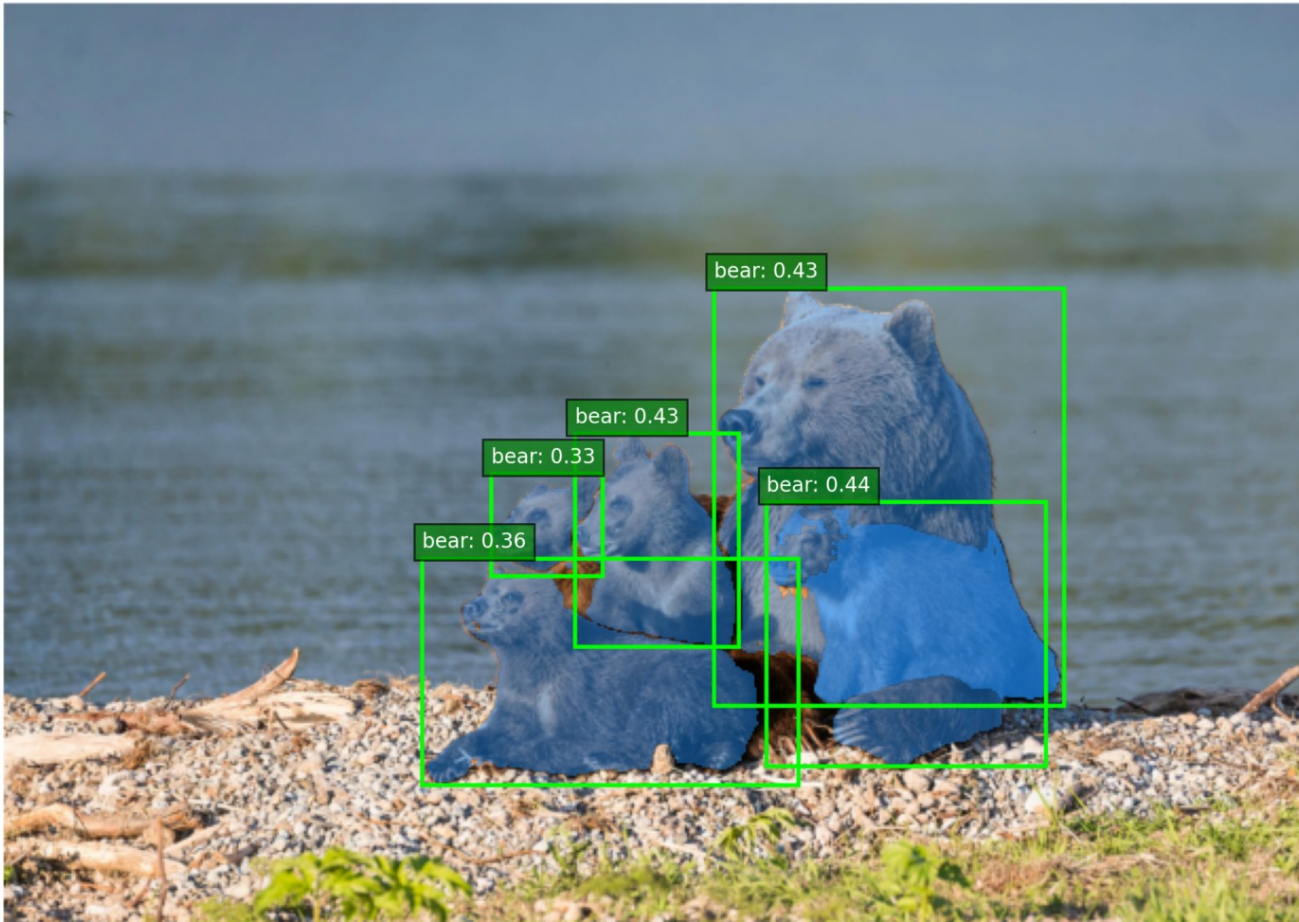


`small animal .` -> lower confidence and wrong semantics

9. Crowded bears: tuned failure-analysis case

Goal: five visible bears -> five final detections, without a false sixth object.

Text-to-mask result: bear .



Final setting

prompt	bear .
box threshold	0.25
text threshold	0.30
NMS IoU	0.35
max detections	8
final detections	5

0.399

mean box score

0.971

mean mask score

Interpretation: moderate detection confidence is expected in an overlapping scene; SAM still produces stable masks once boxes are selected.

10. Proxy metrics summary

No labeled benchmark is used; evaluation combines visual inspection and pipeline-level scores.

Case	After NMS	Mean box	Mean mask	Takeaway
Crowded bears	5	0.399	0.971	overlap bottleneck
Bear cub prompt	4	0.425	0.958	prompt sensitivity
Bus + people	5	0.709	0.962	strong success case
Clear person	2	0.654	0.987	stable simple case
Weak prompt	5	0.286	0.961	semantic mismatch

Box score = detector/localization confidence

Mask score = SAM confidence for selected boxes

Detection is the main bottleneck

11. What was implemented

Individual contribution: the value is a working, reproducible application pipeline.

Reusable pipeline

model loading, detection, NMS, SAM segmentation

Interactive demo

Streamlit app with controls and output table

Colab experiments

sanity check + batch gallery on T4 GPU

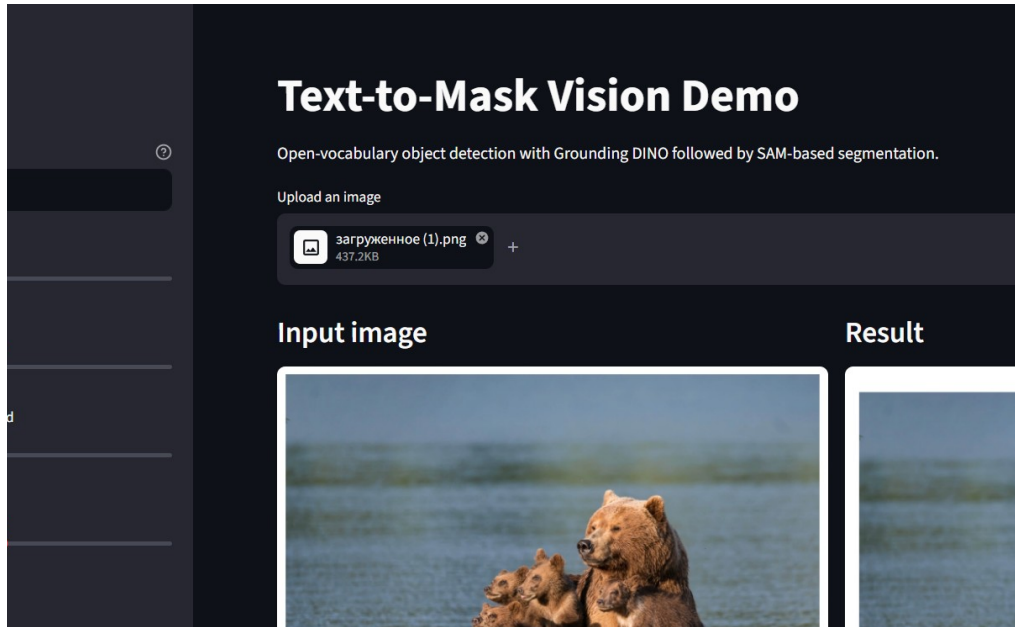
Evaluation artifacts

CSV, figures, report, slides and checks

Not trained from scratch: the project demonstrates integration and analysis of modern pretrained CV foundation models.

12. Conclusion and defense message

A practical, reproducible text-to-mask system built from pretrained vision foundation models.



Final result

- end-to-end text-to-mask pipeline
- interactive Streamlit demo
- qualitative gallery with success and failure cases
- clear limitations and future work

Future work: labeled benchmark, OWL-ViT comparison, latency measurements, mask export, GPU deployment.

Questions?